CHM 120

Laboratory Safety

Final Report

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Test Your Knowledge

Match each term with the best description.

Description		Term
Tightly woven fabric used to smother and extinguish a fire	1	Fire blanket
Consists of absorbent material that can be ringed around a chemical spill until the spill can be neutralized	2	Spill containment kit
Device used to control small fires in an emergency situation	3	Fire extinguisher
Provides chemical, physical, health, and safety information regarding chemical reagents and supplies	4	Safety Data Sheet
Area containing an exhaust fan to expel noxious fumes from a laboratory	5	Fume hood

Label each image with the correct term.

1 Eye-wash station

2 Safety shower

3 First aid kit

4 Burner fuel

5 Fire extinguisher

Categorize each statement as true or false.

True	False
1	2
Always wear safety goggles, long shirts and pants, closed-toe shoes, and gloves when conducting experiments.	Conduct experiments in a small, closed-off room, with no ventilation or water.
Neutralize acid spills using sodium bicarbonate.	Induce vomiting and drink copious amounts of water if chemicals are ingested.
Paper towels and water are a good substitute for a spill containment kit.	

Exploration

For the most part, items in the lab kit are safe for use around children and pets.

- ○ True
- ◉ False

The laboratory kit contains _____.

- ○ small items that can be choking hazards
- ○ chemicals that can cause burns
- ○ sharp objects that can cause cuts
- ◉ All of the above

Eating, drinking, and smoking during an experiment are never allowed.

- ◉ True
- ○ False

To better protect the body from chemical spills, wear _____.

- long-sleeved shirts
- full-length pants
- closed-toe shoes
- All of the above

At home, a _____ is the best substitute for a laboratory safety shower.

- water can
- bottle of water
- kitchen faucet
- bathroom shower

A Safety Data Sheet contains information about first-aid procedures, required protective equipment, and health effects.

- True
- False

After using loops, swabs, toothpicks, spreaders or any other equipment to transfer cultures, immerse the contaminated equipment in ______.

- pure bleach
- a strong acid
- tap water
- All of the above

Burns can occur when skin comes in contact with _____.

- acids
- bases
- acids and bases

Most pieces of glassware are designed to be heated.

- ○ True
- ◉ False

To test the smell of a substance, partially fill your lungs with air and, while standing slightly back from the fumes, use your hand to waft the odors gently toward your nose.

- ◉ True
- ○ False

If any chemical in the kit has been ingested, you should _____.

- ○ call the Poison Control Center
- ○ seek a physician
- ◉ both call the Poison Control Center and seek a physician

Exercise 1

Data Table 1: Lab Safety Equipment Alternatives

Shower or Sink	Paper Towels	Well-Ventilated Area	No Substitution
Bathroom shower/sink	napkins or disposable towels	An office or kitchen	

Exercise 2

Data Table 2: Sodium Hypochlorite SDS information

Items	SDS Information
Physical State	clear, light yellow-green liquid
Route of Exposure & Symptoms	irritation, nausea, headache, shortness of breath
Protective Equipment	NIOSH-approved respiratory protection/breathing apparatus
First Aid Procedures	if enhaled: move person to fresh air; seek medical attention if needed if skin contacts: remove clothing, wash the area, seek medical attention of needed if eyes contact: rinse/flush for 15/20 minutes; immediately get medical assistance if swallowed: rinse mouth, drinks sips of water, seek medical attention if needed
Fire-fighting Measures	Extinguishing Media; Special Hazards Arising From Substance or Mixture
Chemical Reactivity	nonreactive under normal conditions
Safe Storage	providfe ventilation for containers; avoid storage near extreme heat; store away from oxidizing agents; store in cool dry conditions
Safe Disposal	consult federal state/provincial and local regulations regarding the proper disposal of waste; neutralize with dilute acid solutions
Environmental/ Ecotoxicity	

	prevent from reaching drains, sewers, waterways; readily degradable in the environment
Spill Cleanup Procedures	collect liquid and dilute with water; neutrualize with dilute acid solutions;decant water to drian with excess water; absorb with suitable material; dispose remaining solid as normal refuse; always obey local regulations

Exercise 3

What are small-scale techniques?

B I U 14 Word(s)

They are small quantities of chemicals that contribute to the safety of the experiments.

List 5 precautions that must be taken before beginning an experiment.

B I U 31 Word(s)

Wear long, tighter clothing, tie back hair if long, wash hands, be sure to wear lab equipment including gloves and goggles, and read all information and instructions pertaining to the lab.

As indicated in the video, what common substance can be used to neutralize a spilled acid?

B I U 2 Word(s)

Baking Soda

According to the video, why should a used chemical container never be refilled?

B I U 10 Word(s)

Used chemicals can be contaminated and can give unexpected reactions.

Certain glass objects are not meant to be heated and could shatter if exposed to a heat source. What two examples of heat-sensitive glassware are given in the video?

B I U 3 Word(s)

graduated-cylinders and flasks

Exercise 4

Data Table 3: Part 1 and Part 2 of the Safety Contract

Part	Agreement
Part 1: It is impossible to control students' use of HOL Science products, related kits, and students' work environments. The author(s) of HOL Science content, the instructors and institutions that adopt it, and Hands-On Labs, Inc. - the publisher and producer of HOL Science - authorize the use of these educational products only on the express condition that the purchasers and users accept full and complete responsibility for all and any liability related to their use of same. Please review this document several times until you are certain you understand and will fully abide by its terms.	I agree: X
Part 2: I am a responsible adult who has read, understands, and agrees to fully abide by all safety precautions prescribed in HOL Science labs for laboratory work and for the use of HOL Science kits. Accordingly, I recognize the inherent hazards associated with science experimentation; I will always experiment in a safe and prudent manner; and I unconditionally accept full and complete responsibility for any and all liability related to my purchase and/or use of a science HOL Science kit or any other science products or materials provided by Hands-On Labs, INC.	I agree: X

Competency Review

Chemicals in a lab kit may potentially cause _____ if mishandled or consumed.

- burns
- serious illness
- death
- All of the above

When conducting experiments that include vigorous exercise, consult a physician first or ask a partner to assist in participating if you are unable to do so.

- True
- False

The _____ should be contacted immediately if you or anyone accidentally consumes or comes in contact with a potentially toxic substance.

- National Chemical Toxicity Center
- National Poison Control Center
- American Poison Chemical Society
- American Toxic Chemical Center

At the conclusion of experiments involving growing bacterial cultures, _____ should be added to the plates or tubes and everything should be soaked for _____.

- pure acid; 1 hour
- a base; 30 minutes
- pure bleach; 2 hours
- pure bleach; 15 minutes

When working with acids, always _____ to avoid chemical splattering.

- (selected) add acid to water
- add water to acid

Acid spills can be neutralized by adding _____.

- water
- (selected) baking soda
- more acid
- distilled water
- All of the above

Graduated cylinders and volumetric flasks are designed to be heated.

- True
- (selected) False

A _____ can be used to douse a fire.

- fire extinguisher
- safety shower
- fire blanket
- fire blanket and fire extinguisher
- (selected) All of the above

Chemicals may be returned to their containers after they have been dispensed.

- True
- (selected) False

Normal eyeglasses may be worn to substitute for safety goggles.

- True
- False

A _____ expels noxious fumes from a laboratory.

- fire blanket
- fume hood
- spill containment kit
- All of the above

A sink faucet or hand-held shower wand are appropriate substitutes for an eyewash station.

- True
- False

A Safety Data Sheet provides basic information about _____.

- health effects
- toxicity
- physical data
- safe disposal
- health effects and toxicity
- All of the above

First-aid kits are best substituted by bandages and alcohol.

- True
- False

Extension Questions

Use the Internet to look up the SDS for hydrochloric acid and answer the following questions:

a. List the potential acute health effects.

b. List the potential chronic health effects.

c. Identify the first aid measures for ingestion.

d. Record the flammability of the product.

e. Identify 3 chemicals that produce a potentially dangerous reaction with hydrochloric acid.

f. Describe how to handle small spills.

g. List the personal protection when working with the chemical.

(a) Inhalation can cause irritation to nose and upper respiratory tract, ulceration, coughing, chest tightness and shortness of breath.

(b) Tachypnoea, pulmonary oedema and suffocation

(c) If Inhaled: move individual to fresh air and loosen clothing for comfortability; seek medical attention if necessary
After Skin Contact: wash affected area with soap and water; quickly remove contaminated clothing and shoes; rinse with plenty of water for at least 15 minutes; immediately seek medical attention
After Eye Contact: protect unexposed eye; flush thoroughly for at least `15 minutes; remove contact lenses while rinsing; continue to rinse eyes while on the way to the hospital
After Swallowing: rise mouth; do not induce vomiting; have individual drinks sips of water; immediately seek medical attention

(d) non combustible

(e) hydrogen, chlorine, carbon

(f) use water to dilute substance; mop up spill; use sand or something dry to absorb substance spill; dispose of appropriately

(g) ensure adequate ventilation; also ensure that air-handling systems are operational; wear protective eyewear, gloves, and clothing